Artificial Neural Networks

Investigating the Effect of Network Depth and Width on Prediction Performance

Machine Learning Tutorial | Postgraduate Level

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Github

Abstract: Which is better: adding more layers or adding more neurons? This tutorial gives empirical answers based on the Forest Cover Type dataset, 581,012 samples, 54 features, 7 classes of forests with a 103:1 imbalance between classes. It compares two ANN architectures: a Deep Network (10 layers × 64 neurons, BatchNorm), and a Wide Network (2 layers × 512 neurons). The Wide Network has a test accuracy of 87.53% as compared to the Deep Network, which has a test accuracy of 70.76%- a difference of 16.8. One of the most important conclusions is that BatchNorm is vital to the training of 10-layer networks, and class weights will be effective to address the extreme imbalance. This is because Elevation is identified as the most prevalent predictor in SHAP analysis, and this confirms ecological reality. The palette of all figures is colourblind-friendly (Wong, 2011).

1. What Problem Are We Solving?

Mapping Forest cover type based on cartographic measurements is a practical environmental classification problem whose applications are in direct forest management, conservation planning, and ecology. Each observation is an area of a 30m x 30m forest patch in the Roosevelt National Forest, Colorado, which was characterized by terrain and soil variables. The assignment is to guess what type of tree species is dominant there, among 7 different species.

This issue gives an ideal experimental design to test ANN architecture since it is large-scale (581,012 samples), high-dimensional (54 features), multi-class (7 categories), severely imbalanced (103:1 ratio), and an ecological ground truth known against which model explanations can be evaluated. In contrast to the MNIST or CIFAR-10 benchmarks, architectural choices in this case do impact the minority class detection.

What You Will Learn

1. The design and comparison of Deep vs Wide ANN architecture on real tabular data.
2. Why BatchNormalization is necessary to train networks that are more than 5 layers deep.
3. How to deal with extreme class imbalance with class weights at scale.
4. How the SHAP feature importance can be interpreted to explain the model.
5. Why the latency of inference is important outside the accuracy of deployment-level choices.

2. What Does the Data Look Like?

The data are available as the Forest Cover Type (Blackard & Dean, 1999), which is free and available from the UCI Machine Learning Repository, and also through sklearn's built-in loader. It has 581,012 samples and 54 input features, 10 continuously valued cartographic (Elevation, Slope, Aspect, etc.), and 44 binary features (soil and wilderness area).

Property	Detail
Total Samples	581,012
Features	54(10 continuous + 44 binary indicators)
Target Classes	7 cover Types (tree species)
Missing Values	None
Source	UCI Machine Learning Repository / sklearn.datasets

Table 1: Forest Cover Type Dataset Summary

Note: Lodgepole Pine has a percentage of 48.8 out of 100, and Cottonwood/ Willow only has 0.5. Otherwise, models default to the majority classes. This is handled with class weights, which are more effective than SMOTE, given the 581,012 samples.

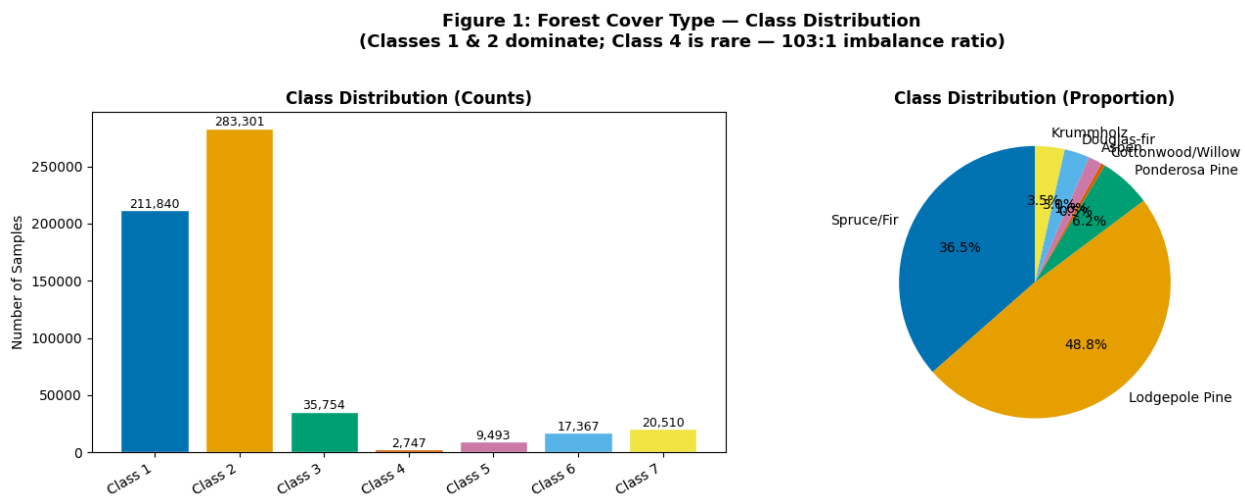

Figure 1: Distribution of forest cover types. Classes 1 (Spruce/ Fir, 36.5%) and 2 (Lodgepole Pine, 48.8%) prevail entirely. Class 4 (Cottonwood/Willow) is virtually invisible at 0.5% - this is the reason why class weights are used in training.

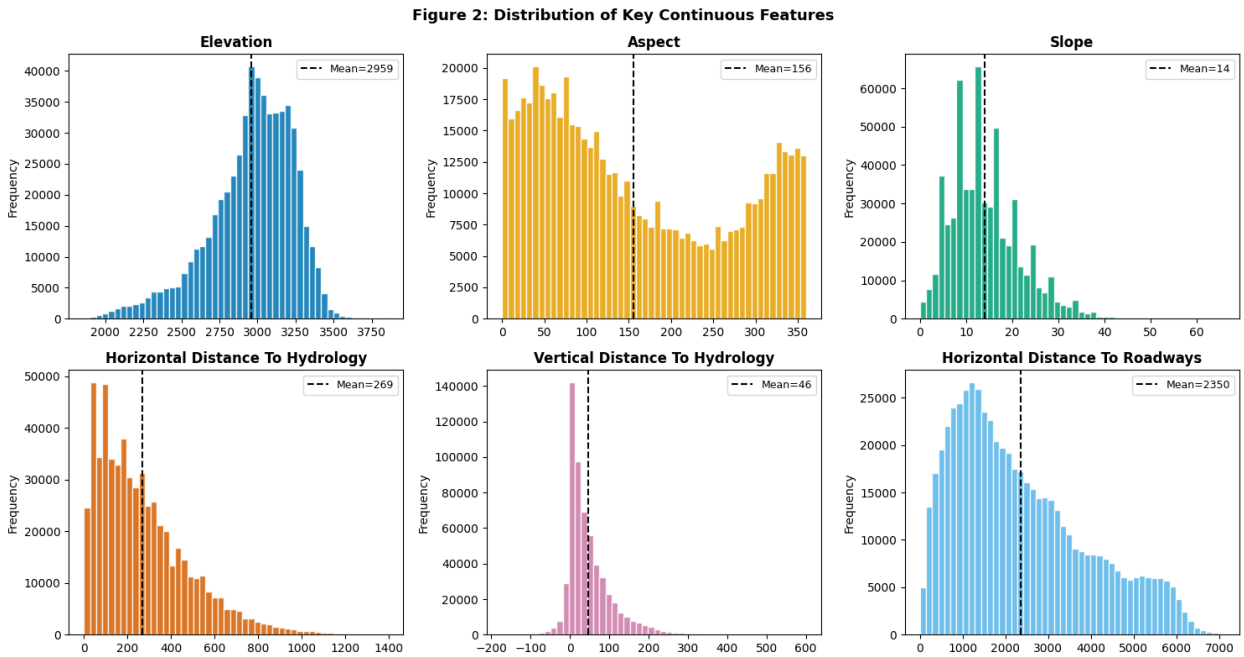

Figure 2: Distributions of the 6 key continuous features. Elevation is approximately normally distributed around 2,959m. Aspect shows a bimodal pattern (north and south-facing slopes). Vertical Distance to Hydrology is heavily right-skewed, requiring

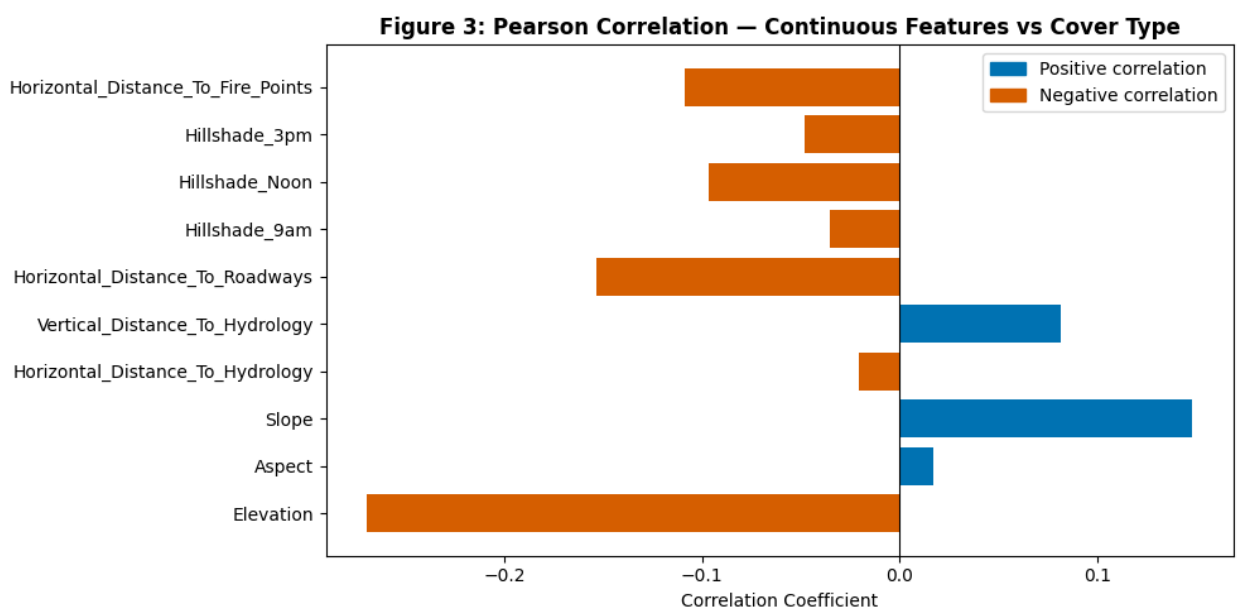

Figure 3: Pearson correlations of the continuous features with cover type. The negative correlation is the most significant with elevation (−0.24), which proves a high altitude as the main ecological driver. The

positive correlation is the strongest between slope and slope (r= +0.15). SHAP analysis will confirm and expand these linear signals in Section 6

3. How Did We Build and Train the Models?

Two ANN architectures that are based on completely different design philosophies are compared. They both employ the same training conditions to isolate the role of architecture.

Preprocessing

1. Target encoding: One-hot vectors (7 columns) of class labels 1-7.
2. Train/test split: 80/20 stratified - maintaining ratios of the classes in both sets.
3. Feature scaling: StandardScaler on the 10 continuous features only (binary indicators are not changed)
4. Class weights: calculated with computeclassweight ('balanced') of sklearn - more loss penalty on rare classes.

Architecture	Hidden Layers	Neuron/Layer	Techniques	Params
Deep Network	10	64	BatchNorm+ Dropout (20%)	42,695
Wide Network	2	512	Dropout (20%)	294,407

Table 2: Architecture Comparison Deep vs Wide ANN

```python
def build_deep_model(input_dim=54, n_classes=7):
    model = Sequential(name='Deep_ANN')
    model.add(layers.Input(shape=(input_dim,)))

    for i in range(10):
        model.add(Dense(64, activation='relu', name=f'dense_{i+1}'))
        model.add(BatchNormalization(name=f'bn_{i+1}'))
        model.add(Dropout(0.2, name=f'drop_{i+1}'))

    model.add(Dense(n_classes, activation='softmax', name='output'))

    model.compile(
        optimizer=keras.optimizers.Adam(learning_rate=0.001),
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )
    return model
```

Building a deep ANN with 10 stacked hidden layers (64 neurons each), Batch Normalization, and 20% Dropouts per layer , Compiled with Adam optimizer And categorical cross-entropy loss for 7- classes

```python
def build_wide_model(input_dim=54, n_classes=7):
    model = Sequential(name='Wide_ANN')
    model.add(layers.Input(shape=(input_dim,)))

    model.add(Dense(512, activation='relu', name='dense_1'))
    model.add(Dropout(0.2, name='drop_1'))

    model.add(Dense(512, activation='relu', name='dense_2'))
    model.add(Dropout(0.2, name='drop_2'))

    model.add(Dense(n_classes, activation='softmax', name='output'))

    model.compile(
        optimizer=keras.optimizers.Adam(learning_rate=0.001),
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )
    return model
```

Building a wide ANN with just 2 hidden layer (512 neurons each) and 20% Dropout per layer trading depth for width

Why BatchNorm for the Deep Network?

After every Dense layer, Batch Normalisation (Ioffe & Szegedy, 2015) re-centres outputs of the layer to mean zero and unit variance. In its absence, gradients disappear exponentially in 10 layers - the model does not learn anything (~14% accuracy, as random as random chance with 7 classes). Using BatchNorm, every 10 layers are faced with consistent gradient values and are trained on meaningful representations. The 2-layer Wide Network does not require BatchNorm - gradient stability also becomes a problem at more than 5 layers.

Training Setting	Value	Reason
Optimizer	Adam (lr=0.01)	Adaptive learning rates handle mixed feature scales
Loss Function	Categorical cross-entropy	Standard for one-hot multi-class targets
Batch Size	512	Large batches stabilize BatchNorm statistics
Max Epochs	30	Both models ran all 30; neither triggered early stopping
Early Stopping	Patience=5	Restores best weights; monitors loss not accuracy
Class Weights	Balanced	Penalizes minority-class errors proportionally

Table 3: Training Configuration

4. What Did the Results Show?

Metric	Deep Network	Wide Network
Test Accuracy	70.76%	87.53%
Test Loss	0.7180	0.3124
Macro F1-Score	0.60	0.81
Weighted F1-Score	0.72	0.88
Training Epochs	30 (no early stop)	30 (no early stop)
Inference Latency	7.839s	6.052s
Params	42,695	294,407
Minority class recall	0.96 – 0.98	0.97 – 1.00

Table 4: Complete Performance Comparison — Deep vs Wide Network

Loss and Accuracy Curves

The loss curve and the accuracy curve depict radically different learning curves. Deep Network begins with a loss of 1.525 (epoch 1) and decreases gradually throughout all 30 of the epochs, still decreasing at the end, which seems to indicate that it never truly converged within the epochs budget. The Wide Network begins with the initial loss of 0.641 and falls drastically, and learns most of its ability in the first 10 epochs, and then refines. The validation accuracy of both models is always greater than the training accuracy, which is a natural consequence of the weights of the classes, which inflate the weighted training loss compared to the unweighted validation evaluation.

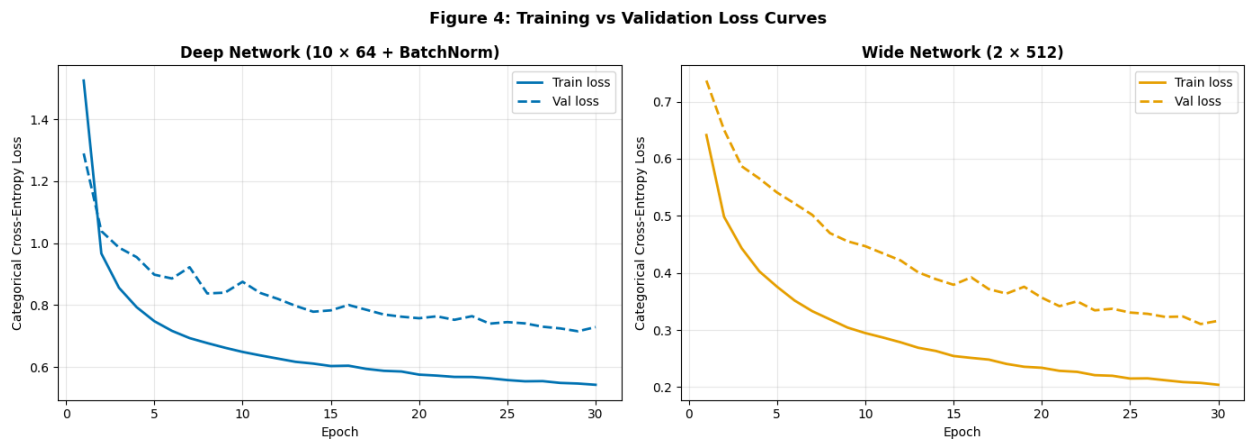

Figure 4: Curves of training and validation loss. The Deep Network loss gradually decreases throughout all 30 epochs and never fully converges. The Wide Network plummets in the initial 10 epochs. Both demonstrate an increase in validation loss over training loss - such is a typical effect of class-weighted training on unweighted validation

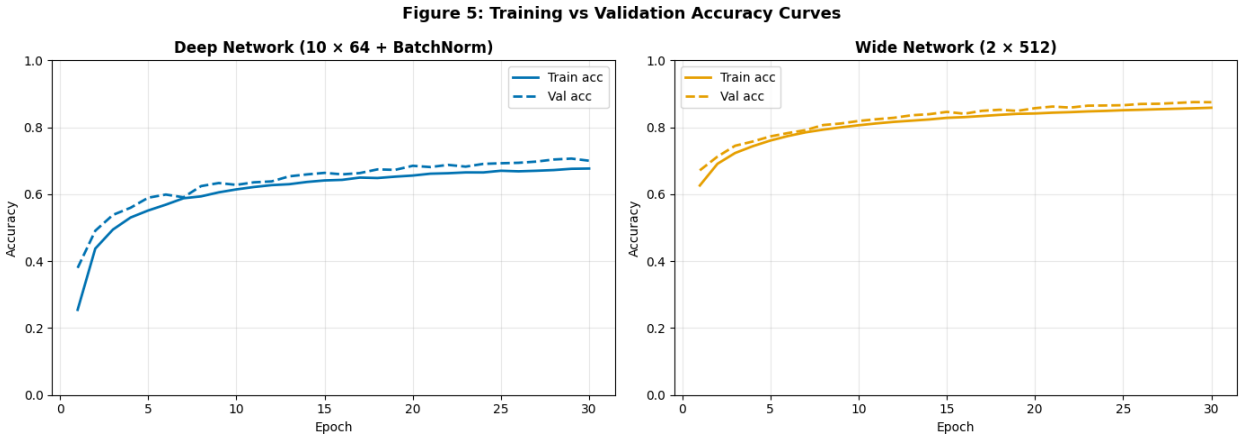

Figure 5: Training vs validation accuracy curves. The Deep Network plateaus around 70% by epoch 15, while the Wide Network continues improving steadily toward 87.5%. The consistent gap confirms the Wide Network's superior representational capacity for this tabular classification task

Where Do the Models Succeed and Fail?

It can be revealed by the confusion matrices what exactly goes wrong in each model. Both of them cope with minority classes very well, a direct result of equalisation of learning pressure, achieved through class weights. In both models, Cottonwood/Willow (0.5% of data) has a recall of 0.97, 0.98, and Krummholz, 0.97-1.00. The biggest weakness of the Deep Network is the majority of classes: Ponderosa Pine recall decreases to only 0.50 on average, i.e., half of all Ponderosa Pine patches are misclassified. It is important to note that 35% of them are forecasted as Douglas-fir - two ecologically close species with overlapping elevations. This ambiguity is much more resolved by the Wide Network (0.89 recall on Ponderosa Pine), which indicates that its larger layers can capture the subtle signaling difference between similar species.

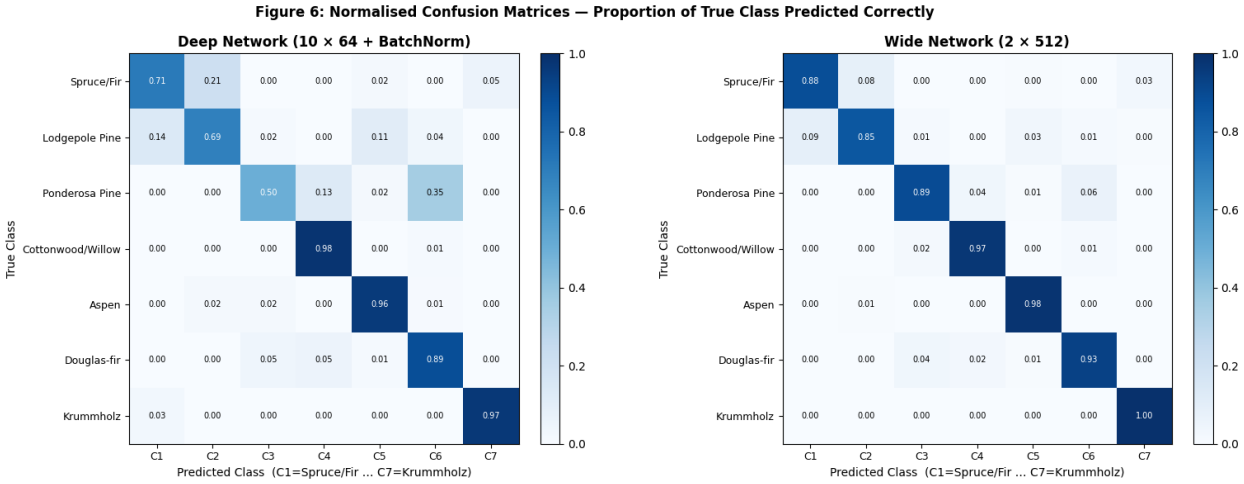

Figure 6: Normalized confusion matrices. The Deep Network struggles with majority classes — Ponderosa Pine (0.50) and Lodgepole Pine (0.69) show the weakest diagonals. The Wide Network achieves consistently strong diagonals across all classes. Both models benefit from class weights on minority classes (Cottonwood/Willow: 0.97–0.98 recall).

Inference Latency

Surprising finding: the Wide Network shows a higher inference speed (6.052s vs 7.839s with 116,203 test samples). Although the Wide Network has 3 times as many parameters, it takes 2 sequential matrix operations to run the 2-layer structure compared to the 10 sequential matrix operations needed to run the 10-layer structure in the Deep Network and 10 sequential BatchNorm operations. In optimized hardware with matrix multiplication, the wider the operation, the more parallelizable the sequential operation is.

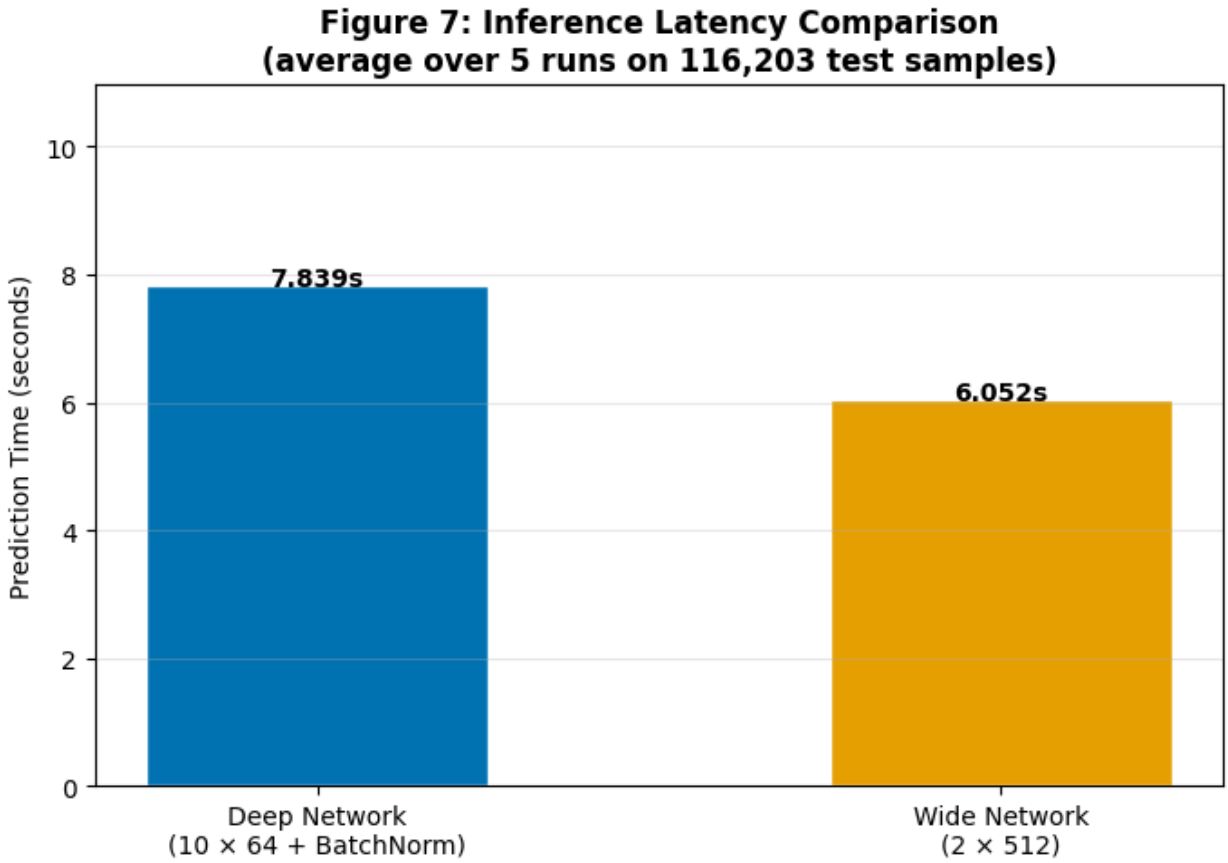

Figure 7: Inference latency over 5 runs on 116,203 test samples. The Wide Network (6.052s) is 1.3× faster than the Deep Network (7.839s) despite having more parameters, because its 2-layer structure requires fewer sequential operations including BatchNorm computations

5. What Drives the Predictions?

The SHAP (Shapley Additive exPlanations) values show what features have the most noticeable impact on the predictions of the Wide Network (Lundberg and Lee, 2017). SHAP is utilized on the Wide Network since a stronger, more useful, and reliable information is obtained when the simpler model is explained. A stratified sample of 500 test points was used to determine values so that the runtime could be manageable.

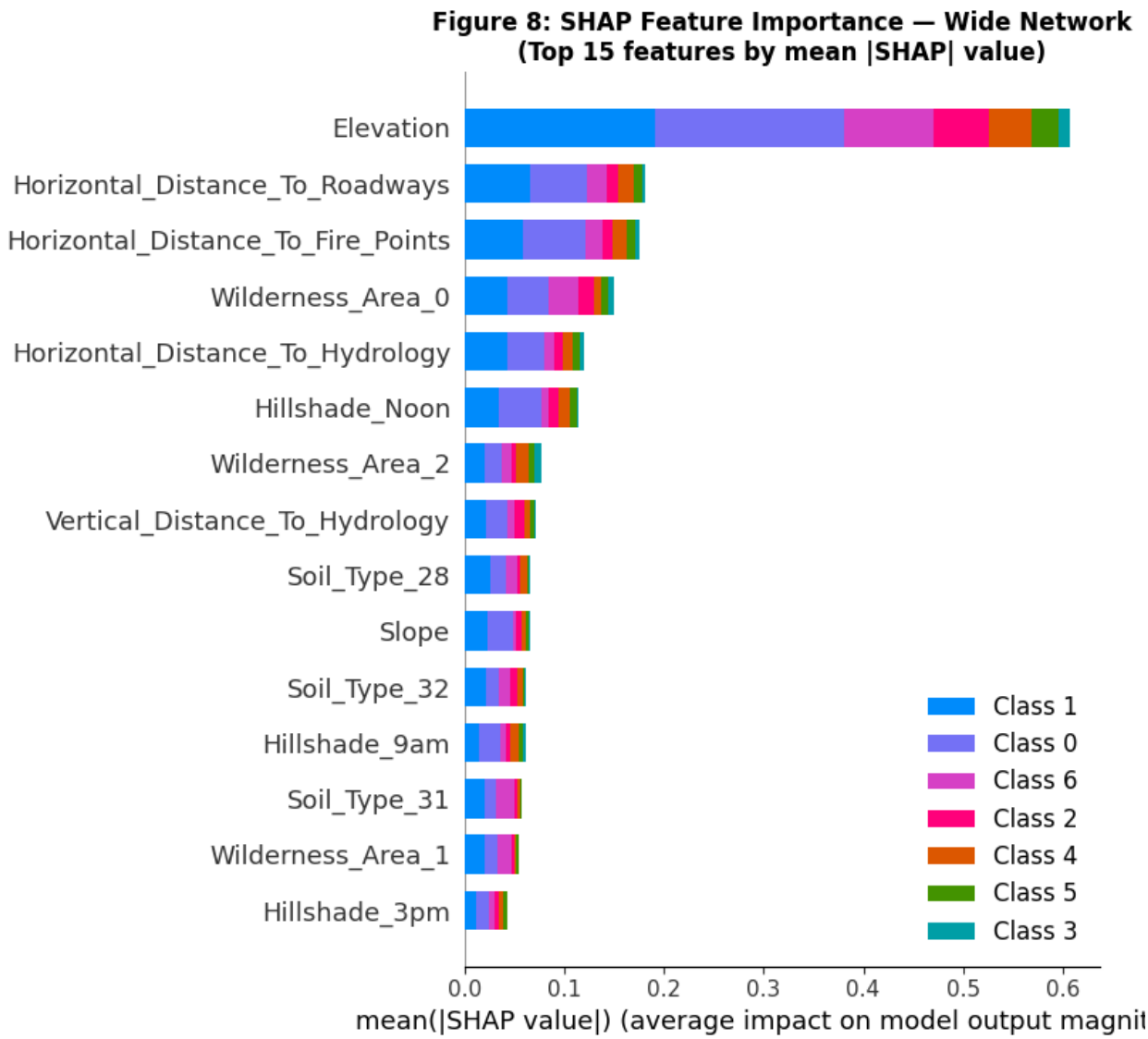

Figure 8: SHAP feature importance for the Wide Network (top 15 features by mean |SHAP| value). Elevation dominates by a large margin, consistent with Figure 3's correlation analysis and ecological reality: altitude is the primary driver of tree species stratification. Horizontal Distance to Roadways and Fire Points rank second and third, reflecting human land-use influence on forest composition

There are two significant validations in the SHAP results. First, the dominance of Elevation proves that the model has acquired ecologically significant patterns as opposed to statistical artefacts. Second, the multi-coloured bars indicate that the effect of Elevation varies among classes. High elevation is a strong predictor of some species (Krummholz, Spruce/Fir), and low elevation predicts other species (Ponderosa Pine, Cottonwood/Willow). This type of direction of influence is only evident in SHAP and would not be evident in straightforward correlation analysis.

6. What Are the Practical Lessons?

- **Depth is not as good as width on tabular data.** The 17-point accuracy difference of the Wide Network demonstrates that 2 large layers are capable of finding more use-case representations among 54 tabular features than are 10 narrow layers. Deep hierarchical learning - which is effective with images - is not the appropriate inductive bias in structured tabular problems.
- **Big unbalanced data sets are well handled with class weights**. SMOTE would be computationally infeasible with 581,012 samples. Class weights obtain the same minority class recall (0.96 -0.98) without the creation of synthetic data. With small datasets, use SMOTE; with large datasets, use class weights.
- **The more parameters, the better the performance**. Wide Network yields 3 times more parameters but converges faster and has better accuracy. The number of parameters is not a good measure of model quality - the architecture form has a much stronger impact.
- **SHAP should be used to verify model understanding.** SHAP established that both models obtained ecologically valid patterns. And, without explainability analysis, we would have no means of differentiating a memorization of training data in a model that simply understood forest ecology.

7. Ethical Considerations

There are actual risks associated with deploying a forest cover type classifier in real environmental decision-making. To begin with, the imbalance in the number of examples of 103:1 classes implies that the model has been trained on much fewer cases of the uncommon species, like Cottonwood/Willow predictions made by the model to support such classes are more likely to be erratic and require careful consideration in conservation planning. Second, the consequences of model errors are asymmetric: a false positive result of identifying a habitat of a species at risk of extinction as a common habitat may result in harmful land use choices. Third, the data were obtained in a single geographic area (Roosevelt National Forest, Colorado) -

extending this model to forests in other areas or climates would introduce systematic bias. Human expert inspection, confidence limits, and periodic retesting of the new field survey should be involved in any deployment.

8. Limitations and Alternative Approaches

Limitations in this experiment:

1. No hyperparameter optimization the learning rate, dropout, and the number of neurons were not tuned manually; grid search or Bayesian optimization can be used to optimize both models better.
2. The issue of class imbalance was not considered - no SMOTE, class weights, or oversampling; one of the reasons the deep network is biased in recall.
3. Findings may not be generalized - the width beats depth observation is particular to this data set; deep structures may be beneficial when dealing with hierarchical or high-dimensional data.
4. SHAP has been calculated on 500 samples a more extensive sample can slightly change the order of features.

Why not XGBoost or Random Forest?

These are the most powerful tabular baselines and tend to be equivalent or better than ANNs with smaller tuning inputs. This tutorial concentrates on ANNs in order to discuss depth vs. breadth, but before making any effort, practitioners are always required to benchmark with tree-based approaches. The complexity of a neural network should be added only in case it can be proven that the network is more effective than other simpler options.

9. Conclusion

It was shown in this tutorial that network width is much faster than depth in tabular forest classification, where the Wide Network (2 × 512) has an accuracy of 87.53% compared to the Deep Network (10 × 64), which has an accuracy of 70.76% given identical conditions during training. The most important lessons are architectural: BatchNorm is needed to make deep networks not have the vanishing gradient effect; class weights are an efficient solution to a 103:1 imbalance without SMOTE; SHAP analysis shows that learned representations are consistent with ecological reality. Senselessly, the Wide Network is also more inference time efficient, since operations on wide matrices are more parallelisable than deep sequential operations with BatchNorm operations.

To practitioners: begin broad on tabular problems, make it deep only with BatchNorm, regularly tackle class imbalance, and use SHAP to confirm that your model has learned something significant, and not just found noise to fit.

References

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